

Work:

Nathan: 2.3,2.4, BE1, BE2, introduction

Yash: 2.5, 2.6, BE6, 5.7

Justin: Use Case, State Diagram, 5.5, 5.6

Christina: scope, 5.2, 5.8, overview

Jasmine: 2.1, system diagram, 5.1, BE7

Muhammad: 5.3, 5.4, BE3, BE4, BE5

1 Introduction

This Software Requirements Specification (SRS) document aims to provide a comprehensive overview of the software requirements for Nutrify, a nutrition application. It will outline the purpose and objectives of Nutrify, define the scope of the application, detail user characteristics, and specify the product requirements. Additionally, it will include use case diagrams to visually represent key system functionalities.

1.1 Purpose

Users post questions about nutrition/macros and the model provides a food recommendation.

1.2 Scope

Nutrify will be a Q&A application dedicated to nutrition and macronutrient goals. Users can post questions about nutritional concerns and receive tailored food recommendations to help them achieve their dietary objectives.

The application will allow users to ask a question, optionally using an account, which will be answered by three different sources: an OpenAI model, a machine learning model, and online sources found by a web scraper. It will report an answer based on the answers given by the sources. Users can also search past questions and filter posts and comments by their preference.

The objective of Nutrify is to provide a niche space for users who may need advice regarding nutrition goals where they can receive a direct response about their unique situation instantly. Another objective of the software is to attract both users and experts, as a larger user base will increase the forum's visibility and credibility.

1.3 Definitions, Acronyms and Abbreviations

N/A

1.4 References

Section found at the end

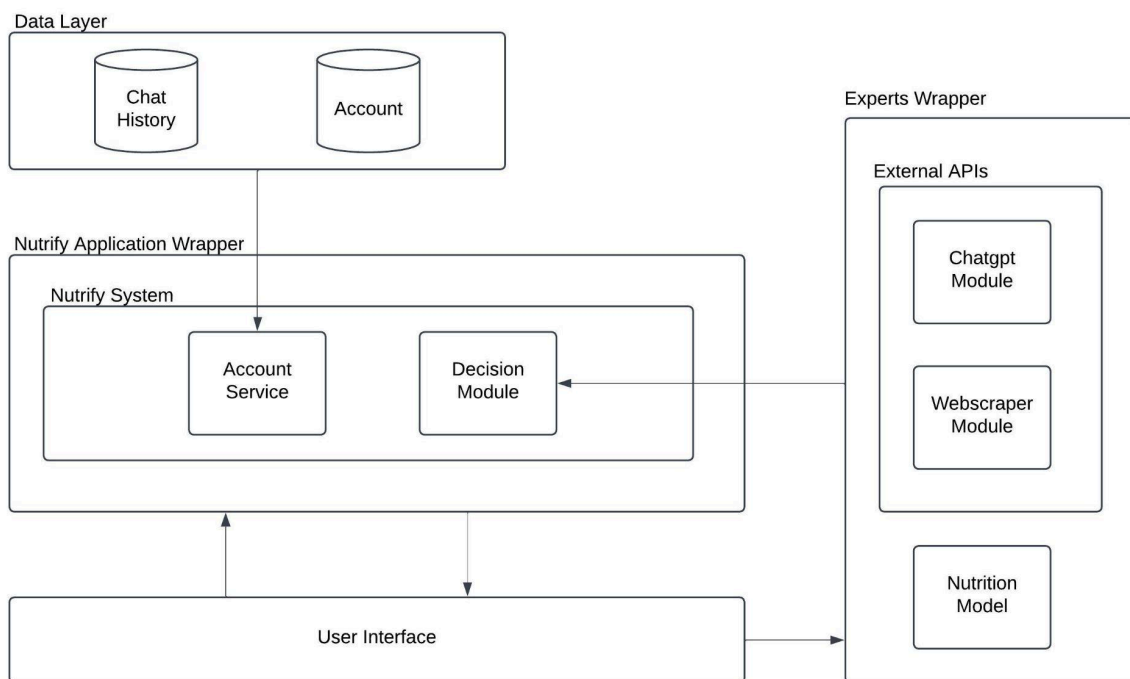
1.5 Overview

Section 2 discusses the overall product description, elaborating on product perspective, product functions, user characteristics, constraints on the product, assumptions and dependencies, and apportioning of requirements. Section 3 contains the use case diagram for the system being developed. Section 4 describes the highlights of functional requirements with use cases, organized by business events and finally, Section 5 describes the product's non-functional requirements.

2 Overall Product Description

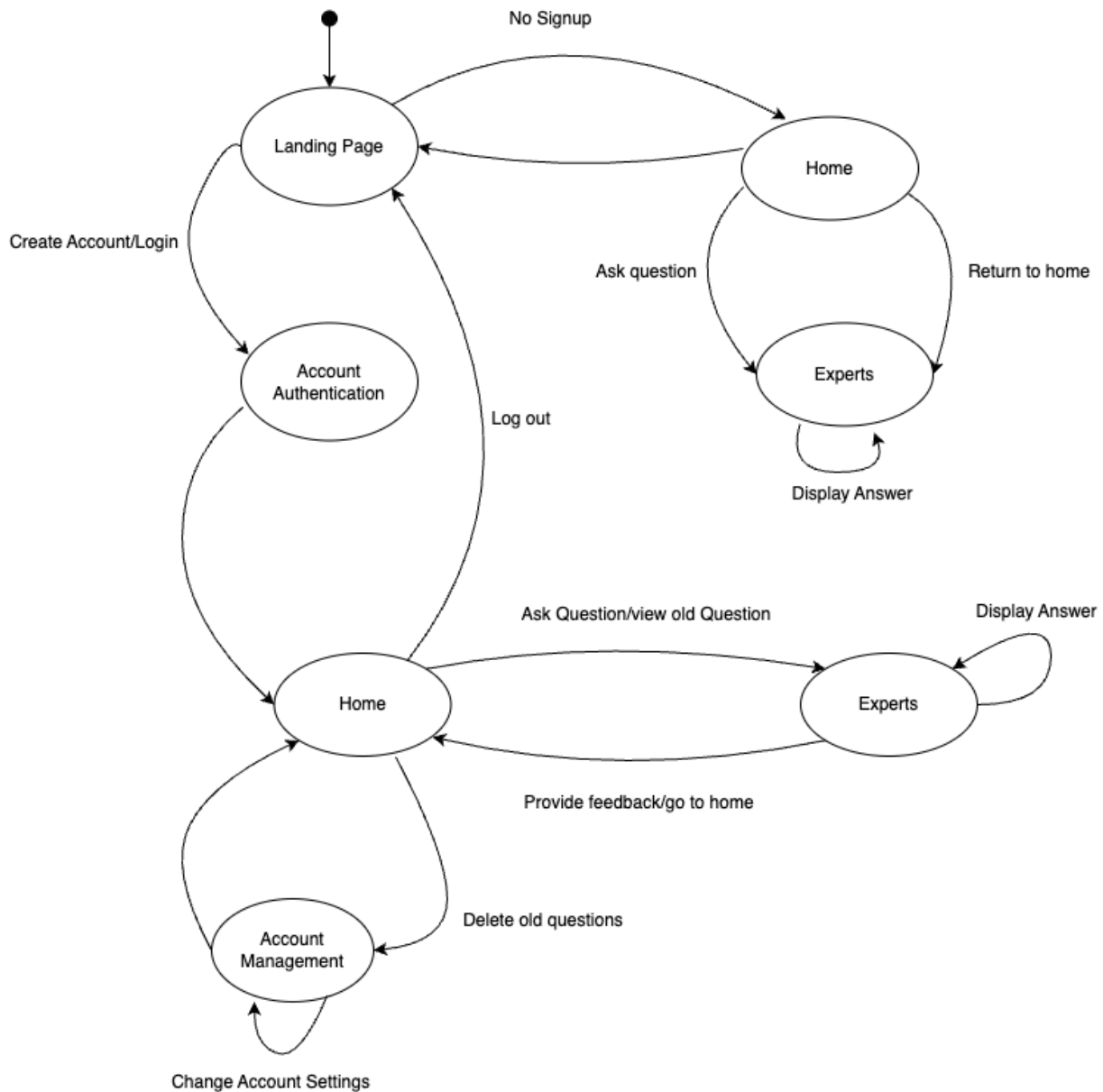
2.1 Product Perspective

Nutrify is designed as an application that uniquely integrates a forum-style architecture with a trio of expert systems to provide personalized nutrition advice. Unlike conventional apps that rely on a single source of information, Nutrify synthesizes recommendations from three distinct experts: an AI large language model(ChatGPT), a web scraper that aggregates up-to-date nutritional data, and a specialized model built for nutrient analysis. The app's decision module then compiles these expert inputs to deliver the best tailored food or meal recommendations to the user based on required nutrients. While Nutrify operates as a standalone product it can act as a potential component within larger health management ecosystems, by bridging AI with holistic nutritional guidance.



2.2 Product Functions

Module(s)	Functions
Account Service	- Create an account (optional) - Delete account - login/authentication
Chat History	- View past chats - Delete past saved chats
Expert 1(AI-1)	- ChatGPT
Expert 2 (Webscraper)	- Answer questions
Expert 3 (Model)	- Use model to answer question
Decision	- Compile the answers from the 3 Experts and decide on best answer



2.3 User Characteristics

The app is meant to be a user friendly app which the user should be:

1. Education Level: Basic Literacy Skills + Highschool level
 - All users should be able to read and write in english and be of atleast highschool level as wellness apps are a bit more on a serious level and we do not want too young users.
2. Technical Experience: Basic skills with a device
 - All users should be able to handle and use an app, this includes navigating the different tabs within the app and viewing comments, liking comments, etc.

3. Experience: Any
 - Since the app is very user friendly, someone who is using the app for the first time should have no issues when navigating through the app

2.4 Constraints

1. Budget

- The overall budget for the project will have a major impact. This is because an app requires a lot of technology, as a result, the developers may be limited to only be able to use a select few.

2. Project Time

- Since there is a planned due date for the overall project, this limits how in depth it can be. Furthermore, it also affects the developing timeline as well as the resources available.

2.5 Assumptions and Dependencies

1. The app will only be used in Canada (1 country)
2. Innovative Feature: You can have your previous prompts saved so that they can be continued at another time.
3. Users have specified any dietary restrictions or allergies when making a prompt
4. The responses provided are just guidance and the final decision is up to the user.
5. The information provided to the user is not a substitute for professional medical advice.
6. Users must have to create an account to allow chat history to be saved.

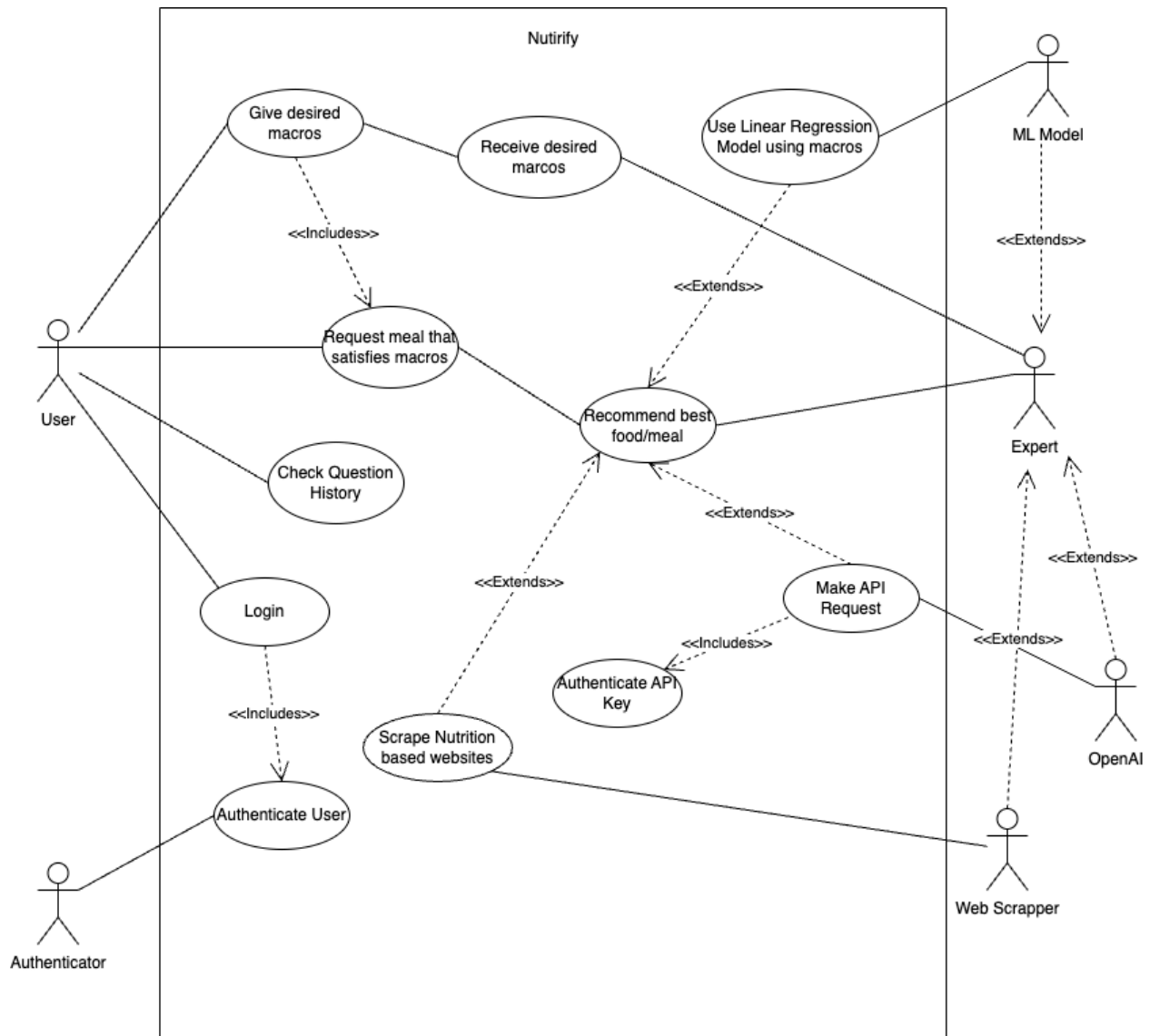
Other assumptions that could require a change to the requirements if they fail to hold:

1. The product should work on the latest available release of the Android operating system.
2. Assume the APIs and databases are fully functional during the operation of the product.
3. Assume the APIs are safe and secure.

2.6 Apportioning of Requirements

1. Further language capabilities
 - a. The software will be developed only in English for the first version of the system.
2. Additional accessibility assistance
 - a. The software should have functions to change text size and color palette to help users who may have color blindness or vision deficits.

3 Use Case Diagram



4 Highlights of Functional Requirements

Viewpoints:

- Users
- Marketing
- Customer support
- Customer Success (Nutrify)

Business events:

- Ask a question
- Delete question history
- Search question history

- Account Sign Up (optional) (referral link)
- Delete account
- Login
- Account Customization

BE1. Ask a question #1

Precondition: There are no current questions that have been asked that answers the user's current question. The user must also have already made an account and already in the app.

VP1. User #1

Main Success Scenario

1. User opens up Nutrify on the their phone
2. System requires the user to input their login information (username and password)
3. User enters login details (te
4. User navigates to and clicks on the option "Ask a question"
5. System displays a new UI to ask a question
6. User puts in the question
7. System verifies that the question is not inappropriate.
8. System outputs an answer

Secondary Scenario

2.2. User does not have an account

2.2.1. System shows option to create an account or login as a guest

7.1. Question is rejected due to inappropriate language, loops back to (6.)

VP2. Marketing #2 (Nutrify)
N/A

VP3. Customer Support #3 (Nutrify)
N/A

VP4. Customer Success #4 (Nutrify)

Main Success Scenario

1. User opens up Nutrify on their phone
2. System requires the user to put in their login information
3. User enters login details
4. User navigates to and clicks on the option "Ask a question"
5. System displays a new UI to ask a question
6. User puts in the question

7. System verifies that the question is not inappropriate.
8. System outputs an answer
9. System asks the user to rate the answer that was provided on a scale from 1-10
10. User gives their feedback
11. System stores the feedback

Secondary Scenario:

- 2.2. User does not have an account
 - 2.2.1. System shows option to create an account or login as a guest
- 7.1. Question is rejected due to inappropriate language, loops back to (6.)
 - 9.1. User does not give feedback, system does not store the feedback.

Global Scenario:

Precondition: There are no current questions that have been asked that answers the user's current question. The user must also have already made an account and already in the app.

1. User opens up Nutrify on their phone
2. System requires the user to put in their login information
3. User enters login details
4. User navigates to and clicks on the option "Ask a question"
5. System displays a new UI to ask a question
6. User puts in the question
7. System verifies that the question is not inappropriate.
8. System outputs an answer
9. System asks the user to rate the answer that was provided on a scale from 1-10
10. User gives their feedback
11. System stores the feedback

Secondary Scenario:

- 2.2. User does not have an account
 - 2.2.1. System shows option to create an account or login as a guest
- 7.1. Question is rejected due to inappropriate language, loops back to (6.)
 - 9.1. User does not give feedback, system does not store the feedback.

BE2. Delete Question History

Precondition: The user has an account created and wants to delete previous questions asked

VPI. User #1

Main Success Scenario:

1. User opens app
2. System requires login information from the user

3. User logs into their account
4. User navigates to the history tab
5. System displays their previously asked questions
6. User can select which questions they have asked previously they want to delete
7. System deletes those questions in the users history

Secondary Scenario:

- 4.1. User has no question history with the app

VP2. Marketing #2 (Nutrify)

N/A

VP3. Customer Support #3 (Nutrify)

N/A

VP4. Customer Success (Nutrify)

N/A

Global Scenario:

Precondition: The user has an account created and wants to delete previous questions asked

Main Success Scenario:

1. User opens app
2. System requires login information from the user
3. User logs into their account
4. User navigates to the history tab
5. System displays their previously asked questions
6. User can select which questions they have asked previously they want to delete
7. System deletes those questions in the users history

Secondary Scenario:

- 4.1. User has no question history with the app

BE3. Search Question History

Precondition: User has an existing account and is logged in

VPI. User #1

Main Success Scenario:

1. User opens app

2. User navigates to the history tab
3. System displays their previously asked questions
4. User enters in filter criteria into search bar to filter questions

Secondary Scenario:

- 3.1. User has no history with the app
- 4.1. User does not filter questions

VP2. Marketing #2 (Nutrify)
N/A

VP3. Customer Support #3 (Nutrify)
N/A

VP4. Customer Success (Nutrify)
N/A

Global Scenario:

Precondition: User has an existing account and is logged in

Main Success Scenario:

1. User opens app
2. User navigates to the history tab
3. System displays their previously asked questions
4. User enters in filter criteria into search bar to filter questions

Secondary Scenario:

- 3.1. User has no history with the app
- 4.1. User does not filter questions

BE4. Create Account

Precondition: User does not have an existing account

VP1. User

Main Success Scenario:

1. User opens app
2. User clicks sign up button
3. User enters in required details
4. System verifies all input fields

Secondary Scenario:

- 4i. User information is invalid
 - 4i.1 System rejects user input information
 - 4i.2 Account creation failed

VP2. Marketing

1. User opens up app through a referral link

VP3. Customer Support

VP4. Customer Success

NA

Global Scenario:

Precondition: User does not have an existing account

Main Success Scenario:

1. User opens app
2. User clicks sign up button
3. User enters in required details
4. System verifies all input fields

Secondary Scenario:

1. User opens up app through a referral link
- 4i. User information is invalid
 - 4i.1 System rejects user input information
 - 4i.2 Account creation failed

BE5. Delete Account

Precondition: User has an existing account and is logged in

VP1. User

Main Success Scenario:

1. User opens app
2. User navigates to account settings
3. User clicks delete account button
4. System prompts user to give reason for account deletion
5. System prompts for confirmation
6. User clicks confirm button
7. System processes account deletion

Secondary Scenario:

- 6i. User clicks decline button
 - 6i.1 Account deletion aborted

VP2. Marketing

NA

VP3. Customer Support

NA

VP4. Customer Success

Global Scenario:

Precondition: User has an existing account and is logged in

Main Success Scenario:

1. User opens app
2. User navigates to account settings
3. User clicks delete account button
4. System prompts user to give reason for account deletion
5. System prompts for confirmation
6. User clicks confirm button
7. System processes account deletion

Secondary Scenario:

- 6i. User clicks decline button
- 6i.1 Account deletion aborted

BE6. Login

Precondition: The user must have already created an account on the platform.

VP 1. User #1

Main Success Scenario:

1. User opens up Nutrify on their phone.
2. System requires User to login and display log-in fields.
3. User enters their login details
4. System authenticates users to see if login details match.
5. User is successfully logged onto the app.

Secondary Scenario:

- 4i. System fails to authenticate the user
- 4i.1 System fails to authenticate the user
- 4i.2 Display login error message, loop back to BE5.VP1.1.

VP 2. Marketing

NA

VP 3. Customer Support

4i.2 System should display error message, and provide a link to customer support to reset password

VP 4. Customer Success

NA

BE7. Account Customization

Precondition: The user must have already created an account on the platform
The user has navigated to the account settings area in the app.

VP 1. User #1

Main Success Scenario:

1. User logs into the Nutrify app with valid credentials.
2. User navigates to the “account settings” page.
3. User clicks the option to “customize account”.

4. System displays the current account settings along with customization options (i.e. dietary preferences/restrictions, personal profile etc.)
5. User updates one or more fields with new information.
6. User confirms changes.
7. System validate input data.
8. System updates the account with the new information.
9. System confirms the update by a success message and displays the updated account details.

Secondary Scenario:

- 4i. Failure to load Account Customization page
 - 4i.1 The system fails to display the customization options
 - 4i.2 The system displays error message prompting to retry.
- 6i. Invalid data entry
 - 6i.1 User enters invalid information in one or more fields.
 - 6i.2 System highlights errors and display error message
 - 6i.3 User corrects informations and resubmit
- 8i. Information update error
 - 8i.1 System fails to save information (i.e. backend error)
 - 8i.2 System displays message to contact customer support.

VP2. Marketing #2 (Nutrify)
N/A

VP3. Customer Support #3 (Nutrify)
4i.2 system should display error message and prompt the user to retry
8i.2 system should display an error message with clear instructions and a link to customer support.

VP4. Customer Success (Nutrify)
N/A

5 Non-Functional Requirements

5.1 Look and Feel Requirements

5.1.1 Appearance Requirements

- **LF-A1:** The app should have a clean and minimalist design

- **Rational:** Ensure the app is visually appealing and easy to navigate. Minimalistic design prevents users from being overwhelmed, especially since WellConnect is a wellness app and users might already be stressed or overwhelmed. Allows user behaviour to be more intuitive and the system to be more easy to operate.
- **LF-A2:** The app should use a consistent color scheme and typography across all screens.
 - **Rationale:** Consistency in color and font allows for a more cohesive user experience and seems more professional. Can reinforce the wellness theme.
- **LF-A3:** The app will have a progress indicator for loading screens or processes that take longer than 2 seconds[].
 - **Rationale:** Progress indicator reassures the user that the app is working even when there isn't immediate feedback. Reduces frustration and confusion where users might be waiting for posts or responses to load,
- **LF-A4:** The app should have a dark and light mode to support user preference
 - **Rationale:** Dark mode reduces eye strain in low-light environments and can save battery life []. Offering both modes ensures accessibility and improves user satisfaction.

5.1.2 Style Requirements

- **LF-S1:** The chatbot should use an empathetic tone in all text content.
 - **Rational:** Aligns with the wellness theme of the app. Make the user feel understood and supported, understand that they are seeking help
- **LF-S2:** Icons and visuals should be intuitive and universally recognizable.
 - **Rational:** Intuitive icons reduce the learning curve for users and make the app easier to navigate. Also mean users from different cultural backgrounds can understand the interface. I.e. arrow pointing up for upvotes and vice versa.
- **LF-S3:** App should adapt to different screen sizes and orientations(portrait and landscape)
 - **Rational:** Ensure the app can be used on a variety of devices with consistent design, This improves accessibility and means users can access the app in a preferred manner to increase satisfaction..
- **LF-S4:** All text is legible (minimum font size?) with sufficient contrast.
 - **Rationale:** Important for accessibility, makes the app readable on most devices, smartphones and tablets.
- **LF-S4:** The logo, color scheme and font used for the app is consistent with the brand
 - **Rationale:** Consistent branding reinforces the app's identity and builds trust with users. Also creates a unified experience.
- **LF-S5:** The app shall have uniform navigation (smooth transitions and animations).

- Rationale: Enhance user experience, seem like a polished app. Simplify the interface and make the system easier to operate.

5.2 Usability and Humanity Requirements

5.2.1 Ease of Use Requirements

- **UH-EOU1:** The app should allow users to provide feedback through an in-app feedback portal.
 - Rationale: User feedback can be used to improve the app in a way that caters to current users. Developers can have a sense of what users want the direction of the app to go towards.
- **UH-EOU2:** The app should have a clear navigation structure that allows users to access key features within three taps.
 - Rationale: Overcomplication of navigation may contribute to a difficult learning curve that can deter users from using the app regularly.
- **UH-EOU3:** The app should have a sufficient user error tolerance such that users can undo mistakes such as misclicks within 2 minutes.
 - Rationale: Users may misclick on or off a page unintentionally. By providing obvious undo and redo buttons, users can easily continue with their intended usage of the app. []

5.2.2 Personalization and Internationalization Requirements

N/A

5.2.3 Learning Requirements

- **UH-L1:** The app should be designed so that new users can use basic functionalities (i.e. asking a question) within 5 minutes
 - Rationale: Use of the app should be intuitive to users, or else users may become frustrated with using the app and cease using it. A difficult learning curve may also deter new users.

5.2.4 Understandability and Politeness Requirements

N/A

5.2.5 Accessibility Requirements

- **UH-A1:** The app should offer the option of changing text size for users with vision difficulties.
 - Rationale: Some users may have difficulty reading small text on their phone. By providing larger text size options, the app ensures a more comfortable and inclusive user experience.
- **UH-A2:** The app should offer the option of changing colour palettes for users with colour blindness.

- Rationale: Some users may not be able to differentiate certain colours appearing on the colour palette. By providing different colour palettes, the app ensures a more comfortable and inclusive user experience.
- **UH-A3:** The app should offer the option of changing the font to a dyslexia-friendly font such as Open Dyslexic.
 - Rationale: Some users may have difficulty reading text due to dyslexia. By providing a dyslexic-friendly font, the app ensures a more comfortable and inclusive user experience.

5.3 Performance Requirements

5.3.1 Speed and Latency Requirements

- PR-SL1. All user interactions and API calls must have an average response time of less than 1 second
Rationale: Fast user interactions are critical in maintaining user engagement and ensuring a smooth user experience [].
- PR-SL2. The overall app startup time must not exceed 2 seconds.
Rationale: Quick startup times help prevent user drop-off and enhance first impressions [].

5.3.2 Safety-Critical Requirements

- PR-SC1. The system must implement safeguards to prevent the display of food recommendations that conflict with users' known dietary restrictions.
Rationale: Ensuring that unsafe recommendations are avoided protects users from potential health risks.
- PR-SC2. The app must make it clear that it is for informational purposes only and should not replace professional dietary advice.
Rationale: Providing disclaimers helps mitigate liability and informs users about the limitations of the recommendations.

5.3.3 Precision or Accuracy Requirements

- PR-PA1. All food recommendations must match the user's specified macronutrient targets within a tolerance of $\pm 5\%$ for each macro
Rationale: High precision in the main functionality of the app is essential for user satisfaction [].

5.3.4 Reliability and Availability Requirements

- PR-RA1. The system must maintain an operational availability of at least 99.9999% (five zeros) over a yearly period (excluding scheduled maintenances)
Rationale: High availability is crucial for user trust and retention [].

- PR-RA2. The system must perform automated daily backups of all user data records to prevent data loss.
Rationale: Reliable data backups ensure continuous service and customer trust in the case of an unfortunate data loss event.

5.3.5 Robustness or Fault-Tolerance Requirements

- PR-RFT1. The system must gracefully handle failures in individual expert components by continuing to function using the others.
Rationale: Robust fault tolerance of the main functionality ensures continuous service and user satisfaction even when part of the system fails.
- PR-RFT2. The system must be able to handle exceptional user inputs.
Rationale: In the case of an exceptional input, the system must fail professionally and avoid any crashes or explicit faults.

5.3.6 Capacity Requirements

- PR-C1. The system must be able to handle 10,000 concurrent users.
Rationale: Based on the C10k problem, Dan Kregel states that modern applications must be able to handle at least 10K users, since often times they need to even go beyond that [].

5.3.7 Scalability or Extensibility Requirements

- PR-SE1. The application architecture must follow modular design principles (eg. SOLID).
Rationale: Modular design promotes easy integration of additional features and minimal maintenance, refactoring, and testing costs [].

5.3.8 Longevity Requirements

N/A

5.4 Operational and Environmental Requirements

5.4.1 Expected Physical Environment

- PR-SE1. The application must maintain consistent performance across diverse network environments, including 3D, 4G, 5G, and Wi-Fi.
Rationale: Reliable operation in variable connectivity is essential for a mobile application.

5.4.2 Requirements for Interfacing with Adjacent Systems

- PR-IA1. The system must seamlessly integrate with external services such as authentication providers, database, and APIs.

- Rationale:** Integration with external systems is crucial since all functionality of the app is tied in said systems one way or another.
- PR-IA12. All communications with external systems must be secured using HTTPS and adhere to RESTful protocols.
- Rationale:** Secure and standardized interfaces such as the ones mentioned protect data integrity and user privacy.

5.4.3 Productization Requirements

N/A

5.4.4 Release Requirements

- PR-R1. The app must comply with all applicable App Store and Google Play Store guidelines.
- Rationale:** Meeting these guidelines is necessary for app deployment and thus public use.
- PR-R2. The app must be compatible with minimum Android OS version 10 and iOS 15.
- Rationale:** These are the minimum recommended versions for both targeted OS's for app performance and reach the largest number of users [1].

5.5 Maintainability and Support Requirements

5.5.1 Maintenance Requirements

- MS-M1. The system must receive at least one software update per week to refine the machine learning model.
- Rationale:** The model should continuously adapt to common macro requirements, evolving alongside Nutrifly's user base.
- MS-M2. The system must receive at least one software update per week to patch bugs and glitches.
- Rationale:** Regular updates help maintain app stability, ensuring a high-quality and enjoyable user experience.

5.5.2 Supportability Requirements

- MS-S1. The system must include an in-app option for accessing customer support.
- Rationale:** Users need a built-in way to request help for navigation issues, general questions, or bug reports. Providing customer support directly within the app ensures a quick and convenient way to get assistance.

5.5.3 Adaptability Requirements

- MS-A1. The system must support the latest Android devices as well as several previous generations to ensure broad user accessibility.
Rationale: Since Nutrify is being developed exclusively for Android, it must be compatible with these devices to ensure a successful launch.

5.6 Security Requirements

5.6.1 Access Requirements

- SR-AC1. The app must request user permission to use their nutrition requests for training machine learning models.
Rationale: To keep the model up to date, the app needs to expand its dataset based on user requests.
- SR-AC2. The app must authenticate user login attempts to verify their identity.
Rationale: This ensures that only the authorized user can access the account, protecting personal data and maintaining security.

5.6.2 Integrity Requirements

- SR-INT1. The app must encrypt all user data during transmission and storage.
Rationale: Encryption helps protect user information from unauthorized access, preventing hackers from stealing sensitive data.
- SR-INT2. The app must support regular data and model checks.
Rationale: To ensure that the data being used for the model does not corrupt and is still valid if any new health information becomes apparent.

5.6.3 Privacy Requirements

- SR-P1. The app must inform users and request their consent before using their information.
Rationale: This aligns with OpenAI's API privacy policy and ensures transparency in data usage[].

5.6.4 Audit Requirements

- SR-AU1. The system must log user requests, questions, ratings, and responses.
Rationale: Keeping detailed logs ensures transparency, helps catch anomalies, and supports security and compliance checks.

5.6.5 Immunity Requirements

- SR-IM1. The system must stay secure and functional even when minor issues or unexpected conditions occur.
Rationale: Fault tolerance helps prevent crashes, data loss, and security risks, keeping the app stable and reliable.

5.7 Cultural and Political Requirements

5.7.1 Cultural Requirements

- **CP-C1:** The system should filter out any terms and icons that may be considered offensive (to any religion, ethnicity, disability, group, etc.), utilizing a list of terms for reference.
 - Rationale: People using the app should feel safe and thus any offensive material should be excluded from the system.
- **CP-C2:** The system shall adapt recommendations based on user's cultural or religious dietary preferences and needs.
 - Rationale: Some cultures follow strict dietary rules and therefore the app must recommend meals that follow these restrictions.

5.7.2 Political Requirements

N/A

5.8 Legal Requirements

5.8.1 Compliance Requirements

- **LR-COMP1:** The app must protect personal information collected from the user.
 - Rationale: According to PIPEDA []
- **LR-COMP2:** The app must inform users of what personal data is being collected and why before and at time of collection.
 - Rationale: According to PIPEDA
- **LR-COMP3:** The app must require the informed consent of users to collect data before personal information is collected or used.
 - Rationale: According to PIPEDA
- **LR-COMP4:** The app must only collect minimum necessary personal information for the function of the app.
 - Rationale: According to PIPEDA
- **LR-COMP5:** The app must only keep personal information for as long as necessary and must only use it for its intended purpose.
 - Rationale: According to PIPEDA

5.8.2 Standard Requirements

- **LR-STD1:** The app must not violate the terms of services of third-party websites used to provide information.
 - Rationale: According to Apple App Review Guidelines []
- **LR-STD2:** The app must not knowingly provide harmful or medically inaccurate fitness advice to users.
 - Rationale: According to Apple App Review Guidelines

- **LR-STD3:** The app must clearly state that responses come from third-party sources or an AI.
 - Rationale: According to Apple App Review Guidelines

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